Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 02 · Part A: Foundations Level 1

Track: Sampling Theory, Aliasing, Interpolation, Bandpass Sampling, Clock Jitter

Prerequisites: Problem 01, trigonometry · **Estimated effort:** 8–12 hours

Sampling, Aliasing and Reconstruction: What a Computer Can and Cannot Know About a Signal

Sampling throws away everything between the samples. The Nyquist–Shannon theorem says when nothing is lost and tells you exactly what goes wrong when it is. Predict aliases by hand, reconstruct signals by sinc interpolation, undersample a radio signal on purpose, and compute how much clock jitter an ADC can tolerate.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/02_sampling_aliasing_and_reconstruction/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis

sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

A flight computer sees the world only at the instants of its sampling clock. If a reaction wheel vibrates at 130 Hz and the gyroscope is read at 100 Hz, the control software does not see a 130 Hz vibration: it sees a 30 Hz wobble that is not there. Aliasing is irreversible; once a signal has been sampled badly, no later processing can tell the true frequency from the false one. The only defences are to sample fast enough, or to remove the offending frequencies *before* sampling with an analog anti-alias filter.

In this problem you simulate the "analog" world on a very dense time grid, sample it, and compare. You will derive the spectral picture of sampling, predict aliased frequencies with a formula, reconstruct a signal from its samples, learn why bandpass signals can legitimately be sampled below twice their highest frequency, and finally quantify the limit that sampling-clock jitter places on the dynamic range of any ADC.

Theoretical Framework

1. Impulse Sampling and Spectral Replication

Sampling $x(t)$ at $t = nT_s$ is multiplication by an impulse train; in the frequency domain it is convolution with an impulse train in frequency, which *replicates* the spectrum:

$$X_s(f) = f_s \sum_{k=-\infty}^{\infty} X(f - kf_s). \tag{1}$$

If $X(f) = 0$ for $|f| > f_{\max}$ and $f_s > 2f_{\max}$, the replicas do not overlap and a low-pass filter recovers $X(f)$ exactly. If $f_s < 2f_{\max}$ they overlap and the overlap cannot be undone.

2. Aliasing

A sinusoid of frequency f sampled at f_s is indistinguishable from one at $f - kf_s$ for every integer k. The apparent frequency in $[0, f_s/2]$ is

$$f_{\text{alias}} = \left| f - f_s \,\text{round}\left(\frac{f}{f_s}\right) \right|. \tag{2}$$

Plotted against f this is a triangle wave ("folding diagram"). Frequencies that are exact multiples of f_s alias to DC and appear as a constant offset.

3. Reconstruction

For a band-limited signal sampled above the Nyquist rate the ideal reconstruction is

$$x(t) = \sum_n x[n]\ \mathrm{sinc}\left(\frac{t - nT_s}{T_s}\right), \qquad \mathrm{sinc}(u) = \frac{\sin \pi u}{\pi u}. \tag{3}$$

It is non-causal and infinite; practical converters use a zero-order hold (ZOH), whose frequency response $H_{\mathrm{ZOH}}(f) = \mathrm{sinc}(f/f_s)\, \mathrm{e}^{-\mathrm{j}\pi f/f_s}$ droops towards the Nyquist frequency and passes images of the spectrum that must be removed by an analog reconstruction filter.

4. Bandpass Sampling

A signal occupying only $f_L < |f| < f_H$ with bandwidth $B = f_H - f_L$ may be sampled below $2f_H$ without aliasing damage provided the replicas interleave without overlapping, i.e. for some integer k with $1 \le k \le \lfloor f_H/B \rfloor$,

$$\frac{2f_H}{k} \le f_s \le \frac{2f_L}{k-1} \qquad \text{(no upper limit for } k = 1\text{)}. \tag{4}$$

Even-numbered replicas are spectrally inverted, which software must undo.

5. Aperture Jitter

If the sampling instants have random error σ_t (RMS), a full-scale sinusoid of frequency f is sampled with an effective amplitude error $2\pi f \sigma_t$ (relative), so the jitter-limited signal-to-noise ratio is

$$\mathrm{SNR}_{\mathrm{jitter}} = -20 \log_{10}(2\pi f\, \sigma_t)\ \ \mathrm{dB}. \tag{5}$$

Data and Simulation Specification

Dense grid	"Analog" time grid at $f_{\mathrm{dense}} = 100\,\mathrm{kHz}$ (or higher); sample by taking every (f_{dense}/f_s)-th point. Always state which frequencies are in Hz.
Test signals	Single tones; the two-tone signal of the session notebook (3 and 7 Hz); a tone at 13 Hz sampled at $f_s \in \{20, 16, 10\}$ Hz; a 130 Hz tone read at 100 Hz.
Bandpass case	Band 70 MHz–80 MHz ($B = 10\,\mathrm{MHz}$), test tone at 74 MHz; candidate $f_s = 20, 23, 25, 28, 30, 33$ MHz.
Jitter	Gaussian timing error $\sigma_t \in \{1, 10, 100\}$ ps, tones at $10, 100, 1000$ MHz, 10^6 samples.
Real data (choose one)	(i) Phyphox accelerometer of a smartphone placed on a running washing machine or fan: record at the highest rate, then deliberately *downsample without filtering* by factors $2, 4, 8$ and observe the aliased peaks. (ii) Kepler long-cadence photometry (cadence 29.4 min, Nyquist 283 µHz) of a pulsating star from the MAST archive via `lightkurve`, compared with the short-cadence (58.9 s) light curve of the same star where available.

Your Task

1. **Spectral replication (Stage 2).** Sample a band-limited signal at $f_s = 100$, 50 and 25 Hz from a dense grid and plot the magnitude spectra on a wide frequency axis (computed from the dense, zero-padded samples). Show the replicas of Eq. (1) and the onset of overlap.

2. **Alias prediction.** Implement Eq. (2). Build the folding diagram (apparent versus true frequency, 0–3f_s). Predict by hand and then confirm numerically the apparent frequency of the 13 Hz tone at $f_s = 20, 16, 10$ Hz and of the 130 Hz tone at 100 Hz. Show that a tone at exactly f_s looks like a constant.
3. **Reconstruction.** Reconstruct the two-tone signal from its samples with Eq. (3), with linear interpolation and with a zero-order hold. Plot the RMS error against the oversampling ratio $f_s/(2f_{\max})$ and against the number of sinc terms kept. Explain the edge effects of a finite record.
4. **Anti-alias filtering.** Add an out-of-band interferer of amplitude 0.5 at 3.7 kHz to a 200 Hz tone and sample at 2 kHz. Show the alias, then design a Butterworth analog pre-filter (simulate it on the dense grid) that suppresses the interferer by at least 40 dB before sampling. State the order needed and the passband droop it causes at 200 Hz.
5. **Zero-order hold.** Compute and verify numerically the ZOH droop at $f_s/4$ and $f_s/2$. Reconstruct a 1 kHz sine from $f_s = 8$ kHz samples with a ZOH and show the spectral images.
6. **Bandpass sampling.** Evaluate Eq. (4) for the 70–80 MHz band and list all valid f_s ranges. Simulate sampling at each candidate rate, locate the apparent frequency of the 74 MHz tone and state for which candidates it is spectrally inverted.
7. **Clock jitter.** Simulate jittered sampling, measure the SNR and compare with Eq. (5) on a plot of SNR versus σ_t for each tone frequency. Convert the SNR into effective bits using ENOB = (SNR − 1.76)/6.02.
8. **Real data.** Analyse your recording or the Kepler data: identify peaks that move when you change the sampling rate (these are aliases) and peaks that do not (these are real). Report which frequencies you trust.

Self-check targets

Apparent frequency of the 13 Hz tone: 7 Hz at $f_s = 20$, 3 Hz at $f_s = 16$ and 3 Hz at $f_s = 10$; the 130 Hz tone read at 100 Hz appears at 30 Hz. For the 70–80 MHz band the valid rates include $f_s = 33$ MHz ($k = 5$) and $f_s = 28$ MHz ($k = 6$ requires $26.7 \le f_s \le 28$) but *not* 25 or 30 MHz. ZOH droop is -0.91 dB at $f_s/4$ and -3.92 dB at $f_s/2$. A 10 MHz tone with $\sigma_t = 10$ ps jitter has SNR ≈ 64 dB (about 10.3 effective bits).

Aerospace Application Link

Reaction wheels, cryocoolers and pumps radiate microvibration at frequencies that depend on their speed. For a gyroscope sampled at $f_s = 100$ Hz compute, for a wheel spinning from 0 to 6000 rpm, the apparent frequency of the fundamental and the first three harmonics, and plot it against wheel speed. Identify the wheel speeds at which a harmonic aliases to *zero frequency* (indistinguishable from a constant gyro bias) and those at which it falls on a structural mode at 12 Hz. Propose a sampling-rate or filter change that removes the problem and compute its cost in data rate.

Deliverables

D1 **Replication notebook:** dense-grid sampling and the three spectra with replicas annotated.

D2 **Alias study:** folding diagram, the predicted versus measured alias table and the DC-alias demonstration.

D3 **Reconstruction study:** sinc, linear and ZOH reconstruction errors versus oversampling ratio, plus the ZOH droop check.

D4 **Anti-alias and bandpass study:** the filter design with attenuation plot, and the table of valid bandpass sampling rates with inversion flags.

D5 **Jitter and real data:** SNR versus jitter plot with the theory curve, and the real-data alias analysis.

D6 **Memo (250 words):** "Why does the reaction-wheel vibration appear as a bias, and what do we change?" Cite at least two external sources.

Bonus (a) Show that non-uniform (random) sampling avoids aliasing at the cost of a noise floor, and compute its spectrum. (b) Estimate the frequency of a tone sampled *below* Nyquist using two different sampling rates (Chinese-remainder idea). (c) Hardware: sample a function generator with a microcontroller ADC at several rates and reproduce your folding diagram.

Learning Resources

Sampling Theory

- Oppenheim, A.V., Schafer, R.W. *Discrete-Time Signal Processing*, Ch. 4 (sampling of continuous-time signals).
- Shannon, C.E. (1949). "Communication in the presence of noise." Proc. IRE 37, 10.
- Nyquist, H. (1928). "Certain topics in telegraph transmission theory." Trans. AIEE 47, 617.
- Vaughan, R.G., Scott, N.L., White, D.R. (1991). "The theory of bandpass sampling." IEEE Trans. Signal Process. 39, 1973.

Converters and Practice

- Kester, W. (ed.) *Data Conversion Handbook*, Analog Devices (sections on aperture jitter and anti-alias filtering); free from the Analog Devices website.
- Lyons, R.G. *Understanding Digital Signal Processing*, Ch. 2 (periodic sampling).

Data and Tools

- Phyphox: `https://phyphox.org`; MAST archive and Lightkurve: `https://archive.stsci.edu`, `https://lightkurve.github.io/lightkurve/`
- SciPy `scipy.signal` (`butter`, `freqs`, `resample_poly`): `https://docs.scipy.org/doc/scipy/reference/signal.html`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.